From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Main), 2014

FULL TEST – IV

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

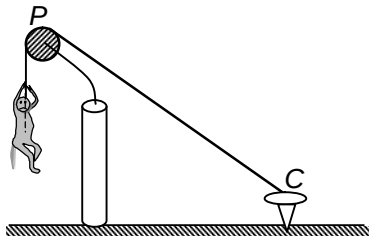
1. Which of the following numbers has maximum number of significant figure?
   (A) 1001 (B) 100.1
   (C) 100.10 (D) 0.001001

2. Two boys are standing at the points A and B on the ground, where AB = a. The boy at B starts running in a direction perpendicular to AB with velocity v_1. The boy at A starts running simultaneously with velocity v and catches the other in time t then t is

   (A) $\frac{a}{\sqrt{v^2 + v_1^2}}$ (B) $\frac{a}{v + v_1}$

   (C) $\frac{a}{v - v_1}$ (D) $\frac{a}{\sqrt{v^2 - v_1^2}}$

3. One end of a massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N with what value of minimum safe acceleration (in ms^{-2}) can a monkey of 60 kg move down on the rope
   (A) 16 (B) 6
   (C) 4 (D) 8



Space for Rough work

4. A particle moves on a rough horizontal ground with some initial velocity say v_0. If $(3/4)^{th}$ of its kinetic energy is lost due to friction in time t_0 then coefficient of friction between the particle and the ground is

(A) $\frac{v_0}{2gt_0}$ (B) $\frac{v_0}{4gt_0}$

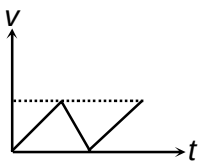
(C) $\frac{3v_0}{4gt_0}$ (D) $\frac{v_0}{gt_0}$

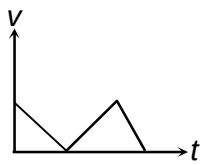
5. A block of mass m is placed on an another rough block of mass M and both are moving horizontally with same acceleration a due to a force which is applied on the lower block, then work done by lower block on the upper block in moving a distance s will be

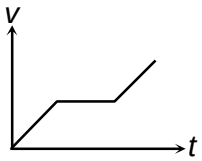
(A) Mas (B) $(m+M)$ as

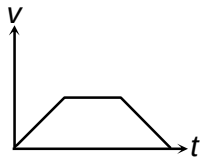
(C) $\frac{M^2}{m}$as (D) mas

6. A ball falls from some height above a horizontal surface. If the collision is elastic, the graph between speed (v) and time (t) upto the second collision looks like

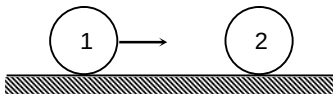
(A) 

(B) 

(C) 

(D) 

7. Ball 1 collides with an another identical ball 2 at rest as shown in figure. For what value of coefficient of restitution e, the velocity of second ball becomes two times that of 1 after collision?



(A) 1/3 (B) 1/2

(C) 1/4 (D) 1/6

Space for Rough work

8. The radius of a planet is R_1 and a satellite revolves around it in a circle of radius R_2. The time period of revolution of satellite is T. Acceleration due to the gravitation of the planet at its surface will be

(A) $\frac{4\pi^2 R_2^3}{T^2 R_1^2}$ (B) $\frac{R_2^3}{4\pi^2 T^2 R_1^2}$

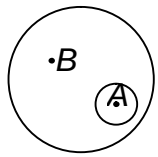
(C) $\frac{4\pi^2 R_1^3}{T^2 R_2^2}$ (D) $\frac{R_1^3}{4\pi^2 T^2 R_2^2}$

9. A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant the velocity of the sphere is 2 m/s. If the coefficient of viscosity of the liquid is $\frac{1}{18\pi}$ N-s/m^2, then velocity of ball will become 0.5 m/s after a time

(A) ln 4 s (B) 2 ln 4 s

(C) 3 ln 4 s (D) 2 ln 2 s

10. There is an air bubble of radius R inside a drop of water of radius 3R. Find the ratio of gauge pressure at point B to that at point A.



(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{3}$ (D) 1

11. A simple pendulum is suspended from the ceiling of an empty box falling in air near earth surface. The total mass of system is M. The box experiences air resistance $\vec{R} = -k\vec{v}$ where v is the velocity of box and k is a positive constant. After some time it is found that period of oscillation of pendulum becomes double the value when it would have suspended from a point on earth. The velocity of box at that moment (take g in air same as on earth's surface)

(A) $\frac{Mg}{4k}$ (B) $\frac{Mg}{k}$

(C) $\frac{Mg}{2k}$ (D) $\frac{2Mg}{k}$

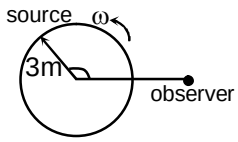
Space for Rough work

12. A uniform vertical electric field E is established in the space between two large parallel plates. A small conducting sphere of mass m is suspended in the field from a string of length L. If the spheres is given a positive charge q and the lower plate is charged positively, the period of oscillation of this pendulum is

(A) $2\pi\sqrt{\frac{L}{g}}$ (B) $2\pi\sqrt{\frac{L}{g+(qE/m)}}$

(C) $2\pi\sqrt{\frac{L}{g-(qE/m)}}$ (D) $2\pi\sqrt{\frac{L}{[g^2+(qE/m)^2]^{1/2}}}$

13. A source is moving on a circle of radius 3 m with constant angular velocity ω = 5 rad/s. If the observer is at a distance 5 m from the centre of circle, the time interval between maximum and minimum frequency received by the observer is



(A) $\frac{\pi}{5}$ (B) $\frac{2}{5}\cos^{-1}\left(\frac{3}{4}\right)$

(C) $\frac{2}{5}\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\frac{2}{5}\sin^{-1}\left(\frac{3}{5}\right)$

14. A l cm long string fixed at both ends, sustains a standing wave such that all the points on the string having displacement amplitude 1 mm (less than maximum amplitude) are separated by d cm. The string is oscillating in its third overtone then

(A) $\frac{l}{d}=2$ (B) $\frac{l}{d}=3$

(C) $\frac{l}{d}=6$ (D) $\frac{l}{d}=8$

15. The air in a open pipe of length 36 cm long is vibrating with 2 nodes and 2 antinodes. The temperature of the air inside the pipe is 51^0C. What is the wavelength of waves produced in air outside the tube where the temperature of air is 16^0C?

(A) 32.1 cm (B) 68 cm

(C) 34 cm (D) 10.2 cm

Space for Rough work

16. A cubical vessel of edge 1 m and total thermal resistance (of its walls) is $\frac{1}{R}$ (where, R is universal gas constant) has a small hole in one of its walls. It is kept in a very big closed chamber whose temperature T_0 remains constant. In the chamber and vessel, a mono-atomic gas is filled at a same constant pressure P_0. At time t = 0, temperature of the gas in the vessel is T_1 (< $2T_0/3$). When temperature of the gas in the vessel becomes 0.8 T_0, rate of change of moles in it will be

(A) $\frac{1}{10}$ (B) $\frac{1}{6}$

(C) $-\frac{1}{10}$ (D) $-\frac{1}{6}$

17. One mole of an ideal gas is enclosed in a cylinder fitted with a frictionless piston and occupies a volume of 1.5 litre at a pressure of 1.2 atm. It is subjected to a process given by equation $T = \alpha V^2, \gamma$ (adiabatic constant) for the gas = 1.5. Choose the wrong statement.

Given $R\alpha = 80$ J mol^{-1} lit^{-2} (R = gas constant and α is constant)

(A) the P-V diagram of the process is a straight line.
(B) the work done by the gas in increasing the volume of the gas to 9 litre is 3150 J.
(C) the change in the internal energy of the gas is 12600 J.
(D) the heat supplied to the gas in the process is 15750 J.

18. A point charge is placed at a distance r from center of a conducting neutral sphere of radius R ($r >$ R). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by [k = $\frac{1}{4\pi\varepsilon_0}$]

(A) kq/r_1 (B) kq/r
(C) $kq/r - kq/r_1$ (D) $-kq/R$

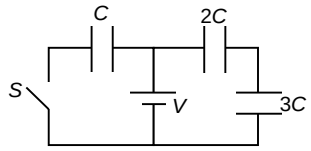
19. A charge q is placed at some distance along the axis of a uniformly charged disc of surface charge density σ. The flux due to the charge q through the disc is ϕ. The electric force on charge q exerted by the disc is

(A) $\sigma\varphi$ (B) $\frac{\sigma\varphi}{4\pi}$

(C) $\frac{\sigma\varphi}{2\pi}$ (D) $\frac{\sigma\varphi}{3\pi}$

Space for Rough work

20. In the given circuit diagram, find the heat generated on closing the switch S. (Initially the capacitor of capacitance C is uncharged)



(A) $\frac{3}{2}CV^2$ (B) CV^2

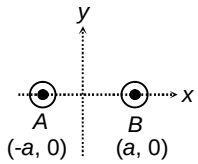
(C) $\frac{1}{2}CV^2$ (D) $2CV^2$

21. A spherical conductor A of radius r is placed concentrically inside a conducting shell B of radius R(R > r). A charge Q is given to A, and then A is joined to B by a metal wire. The charge flowing from A to B will be

(A) $Q\left(\frac{R}{R+r}\right)$ (B) $Q\left(\frac{r}{R+r}\right)$

(C) Q (D) zero

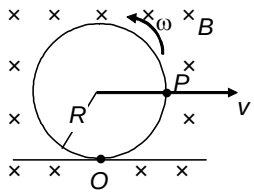
22. Two very long current carrying wires A and B carrying current I_0 (along z-axis) are placed at (–a, 0) and (a, 0) as shown. Find the value of magnetic field at (0, a)



(A) $\frac{\mu_0 I_0}{\sqrt{2}\pi a}$ (B) $\frac{\mu_0 I_0}{2\pi a}$

(C) $\frac{\mu_0 I_0}{4\pi a}$ (D) $\frac{\mu_0 I_0}{2\sqrt{2}\pi a}$

23. A metallic ring of radius R moves in a vertical plane in the presence of a uniform magnetic field B perpendicular to the plane of the ring. At any given instant of time its centre of mass moves with a velocity v while ring rotates in its COM frame with angular velocity ω as shown in the figure. The magnitude of induced e.m.f. between points O and P is

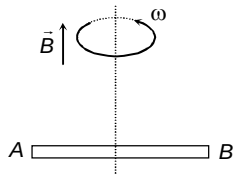


(A) zero (B) $vBR\sqrt{2}$

(C) vBR (D) 2 vBR

Space for Rough work

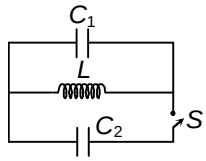
24. A conducting rod of length 2l is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is



(A) $B\omega l^2$ (B) $\frac{1}{2}B\omega l^2$

(C) $\frac{1}{8}B\omega l^2$ (D) zero

25. At a moment (t = 0), when the charge on capacitor C_1 is zero, the switch is closed. If I_0 be the current through inductor at t = 0 , for t > 0 (initially C_2 is uncharged)



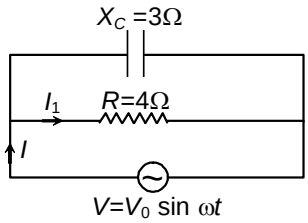
(A) maximum current through inductor equals $I_0/2$.

(B) maximum current through inductor equals $\frac{C_1 I_0}{C_1 + C_2}$.

(C) maximum charge on $C_1 = \frac{C_1 I_0 \sqrt{LC_2}}{C_1 + C_2}$.

(D) maximum charge on $C_1 = C_1 I_0 \sqrt{\frac{L}{C_1 + C_2}}$.

26. A capacitor and resistor are connected with an A.C. source as shown in figure. Reactance of capacitor is $X_C = 3\Omega$ and resistance of resistor is 4Ω. Phase difference between current I and I_1 is $\left[\tan^{-1}\left(\frac{3}{4}\right) = 37^o\right]$



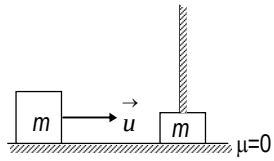
(A) 90° (B) zero

(C) 53º (D) 37º

Space for Rough work

27. In Young's double slit experiment the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same set-up the two slits are source of equal amplitude A and wavelength λ, but are incoherent. The ratio of the intensity of light at the midpoint of the screen in the first case to that is second case is
(A) 1 (B) 2
(D) 3 (D) 4

28. An object of mass m is moving with velocity $\vec{u}$ towards a plane mirror kept on a stand as shown in the figure. The mass of the mirror and stand system is m. A head-on elastic collision takes place between the object and the mirror stand, the velocity of image before and after the collision is



(A) $\vec{u}, 2\vec{u}$ (B) $-\vec{u}, -2\vec{u}$
(C) $-\vec{u}, 2\vec{u}$ (D) $\vec{u}, -2\vec{u}$

29. A silver sphere (work function 4.6 eV) is suspended in a vacuum chamber by an insulating thread. Ultraviolet light of wavelength 0.2 μm strike on the sphere. The maximum electric potential of the sphere will be (hc = 12400 eVÅ)
(A) 4.6 V (B) 6.2 V
(C) 1.6 V (D) 3.2 V

30. If light of wavelength of maximum intensity emitted from a surface at temperature T_1 is used to cause photoelectric emission from a metallic surface, the maximum kinetic energy of the emitted electron is 6 eV, which is 3 times the work function of the metallic surface. If light of wavelength of maximum intensity emitted from a surface at temperature $T_2 (T_2 = 2T_1)$ is used, the maximum kinetic energy of the photoelectrons emitted is
(A) 2 eV (B) 4 eV
(C) 14 eV (D) 18 Ev

Space for Rough work

Chemistry

PART – II

SECTION – A

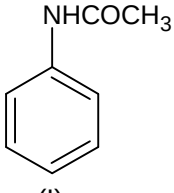
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

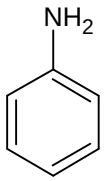
1. The ratio of difference of radius between 3rd and 2nd orbit of H-atom and Li^{+2} ion is:
   (A) 3 : 1 (B) 1 : 1
   (C) 1 : 3 (D) 2 : 3

2. The slag obtained during the extraction of copper pyrites is composed mainly of:
   (A) $MgSiO_3$ (B) $CuSiO_3$
   (C) $FeSiO_3$ (D) $CuFeS_2$

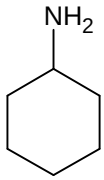
3. Rank in decreasing order of basicity:



(I)



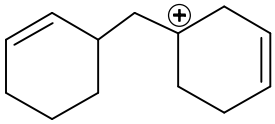
(II)

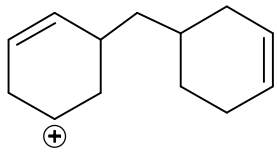


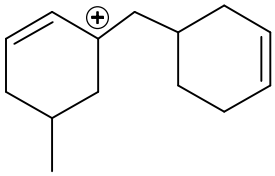
(III)

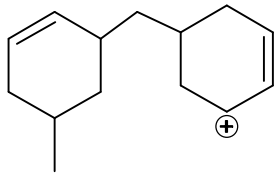
   (A) III, I, II (B) III, II, I
   (C) II, I, III (D) II, III, I

4. Which of the following is the most stabilized carbocation?

(A)


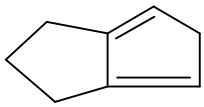
(B)


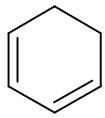
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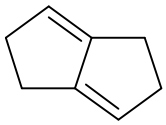
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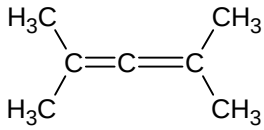
Space for Rough work

5. Which of the diene has highest heat of hydrogenation?

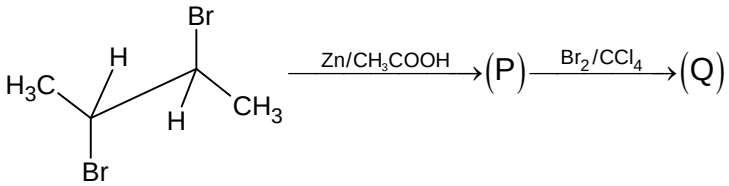
(A) 

(B) 

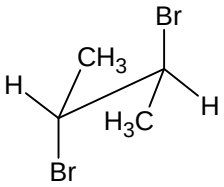
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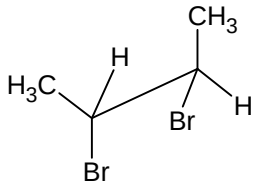
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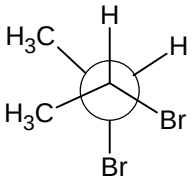
6. If the solubility of calcium phosphate $[Ca_3(PO_4)_2]$ is s mol L^{-1}. The solubility product is equal to
(A) $6s^5$ (B) $72s^5$
(C) $54s^4$ (D) $108s^5$

7. 

Q is major product?

(I) 

(II) 

(III) 

(A) I only (B) II only
(C) III only (D) I, II and III

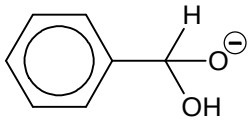
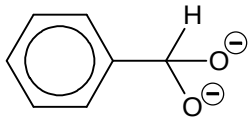
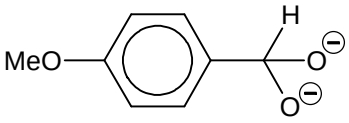
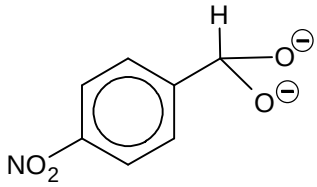
8. The reaction of NH_3 and Cl_2 is as follows

$$aNH_3 + bCl_2 \longrightarrow cNH_4Cl + N_2$$

The coefficient a, b and c are respectively.
(A) 8,4,6 (B) 8, 3, 2
(C) 3, 8, 6 (D) 8, 3, 6

Space for Rough work

9. A crystal made up of A, B and C atoms, A occupy in all the tetrahedral voids and B occupied in octahedral void while 'C' forms face centred cubic. The molecular formula of the compound or crystal is:
(A) AB_2C (B) $A_2B_3C_8$
(C) $A_3B_8C_3$ (D) A_2BC

10. In a Cannizzaro reaction, the intermediate that will be best hydride donor is:
(A) 
(B) 
(C) 
(D) 

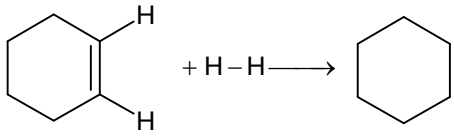
11. The number of revolutions made by an electron in Bohr's 2nd orbit of hydrogen atom in one second is (approx.)
(A) 8.3×10^{14} (B) 6.66×10^{15}
(C) 5.3×10^{6}` (D) none of these

12. Two non reacting gases A and B, having the mole ratio of 3 : 5 in a container, exert a pressure of 8 atm. If B is removed, what would be pressure of A only, T remaining constant.
(A) 3 atm (B) 4 atm
(C) 5 atm (D) None of these

13. A liquid is kept in a closed vessel. If a glass plate with a small hole in kept on top of the liquid surface, what would be the vapour pressure of the liquid in the vessel? (Neglect weight of glass)
(A) More than what would be if the glass plate were removed.
(B) Less than what would be if the glass plate were removed.
(C) Same as what would be if the glass plate were removed.
(D) Can not predicted.

14. Two mole of ideal gas at 300 K is expanded isothermally from an initial volume of 1 litre to 10 litre. The change in internal energy in the process is:
(A) 11.49 kJ (B) 2.73 kcal
(C) zero (D) All are incorrect

Space for Rough work

15. 

Bond energies of C – C, C = C, C – H and H – H are w, x, y and z kJ/mole respectively. What is the value of ΔH^o for above reaction?

(A) 2y – x – z) kJ
(B) (x + z – w – 2y) kJ
(C) (x + z – y) kJ
(D) (x + z – 2y) kJ

16. For a reaction

$3A + B \longrightarrow 2C$

$-\frac{d[A]}{dt} = K[A][B]$, the expression for $\frac{d[C]}{dt}$ will be:

(A) $\frac{2}{3}K[A][B]$
(B) $\frac{3}{2}K[A][B]$
(C) $\frac{1}{3}K[A][B]$
(D) $\frac{1}{2}K[A][B]$

17. For the reaction $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g)$

the pressure of CO_2(g) depends on

(A) Amount of $SrCO_3$(s)
(B) Amount of SrO (s)
(C) Amounts of both $SrCO_3$(s) and SrO(s)
(D) Temperature of the system

18. On increasing temperature ionic product of water increases $\left[H_2O \rightleftharpoons H^+ + OH^-, K_W = 10^{-14} \text{ at } 25^oC\right]$ which of the following statement is correct regarding dissociation of water.

(A) water becomes basic on decreasing the temperature
(B) pH of water increase on increasing the temperature
(C) water will be neutral irrespective of temperature change
(D) pH cannot be predicted on given information

19. On adding few drops of dil. HCl to freshly precipitated ferric hydroxide, a red coloured colloidal solution is obtained. The phenomenon is known as:

(A) protective action
(B) flocculation
(C) coagulation
(D) peptisation

20. When sodium argentocyanide is treated with zinc dust, silver precipitates because

(A) Both Zn^{+2} and Ag^+ ions have d^{10} electronic configuration
(B) Zinc forms a complex readily with cyanide
(C) Zinc is more electropositive than silver
(D) Zinc is less electropositive than silver

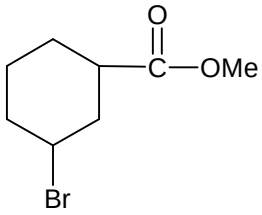
Space for Rough work

21. Aqueous solution of carnalite gives a positive test for
(A) potassium ions only
(B) chloride ions only
(C) magnesium and chloride ions
(D) potassium, magnesium and chloride ions

22. Which compound is formed when excess of KCN is added to an aqueous solution of copper sulphate?
(A) $Cu(CN)_2$
(B) $K[Cu(CN)_2]$
(C) $K_4[Cu(CN)_6]$
(D) $K_3[Cu(CN)_4]$

23. In successive emission of α and β particles, how many α and β particles should be emitted for conversion $_{94}Pu^{241}$ to $_{92}U^{233}$?
(A) $2\alpha, 3\beta$
(B) $3\alpha, 2\beta$
(C) $2\alpha, 2\beta$
(D) none of them

24. The IUPAC name of the following compound

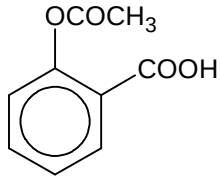


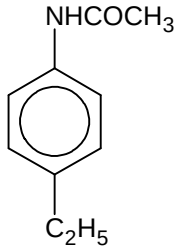
(A) 3-bromo-1-methoxy cyclohexanone
(B) 4-bromo-1-methoxy cyclo hexanone
(C) methyl-3-bromo cyclo hexanoate
(D) methyl-3-bromo cyclo hexane carboxylate

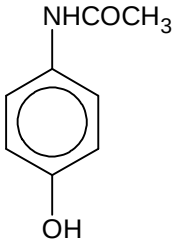
25. The excessive use of phosphates as water softners leads to aquatic pollution called
(A) eutrophication
(B) phsophosylation
(C) deoxygenation
(D) denitrification

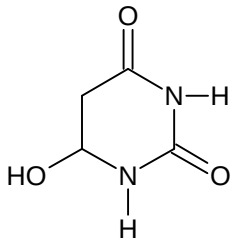
Space for Rough work

26. Paracetamol is a common antipyretic, which is the correct formula among given options?

(A) 

(B) 

(C) 

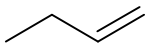
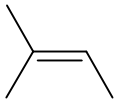
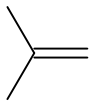
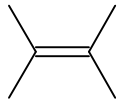
(D) 

27. Consider the following statements about lactose:
1. Lactose on hydrolysis yields equimolar mixture of D-glucose and D-galactose.
2. Lactose is a poly-saccharide.
3. Lactose is a reducing sugar.
4. In lactose, the glycosidic linkage is between C-1 of galactose and C-4 glucose unit.

Which of the statements are correct?

(A) 1 and 3 only
(B) 1 and 4 only
(C) 2 and 3 only
(D) 1, 3 and 4 only

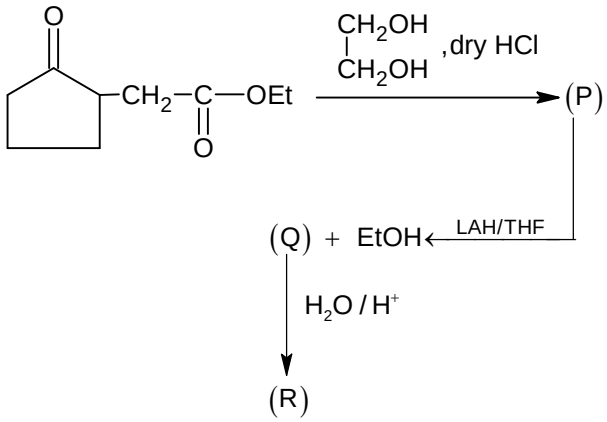
28. The correct order of reactivity of alkenes towards HBr is:

1. 
2. 
3. 
4. 

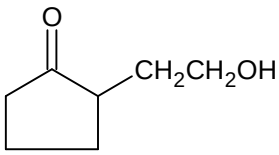
(A) 1 > 2 > 3 > 4
(B) 1 > 3 > 2 > 4
(C) 3 > 2 > 4 > 1
(D) 4 > 2 > 3 > 1

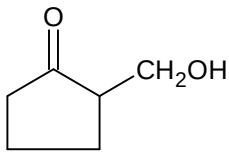
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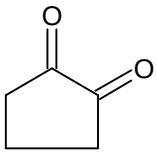
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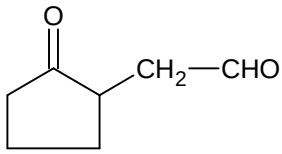


What is the structure of (R)?

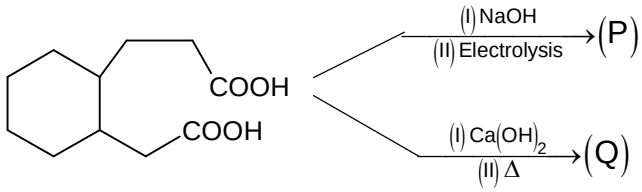
(A) 

(B) 

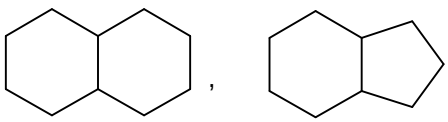
(C) 

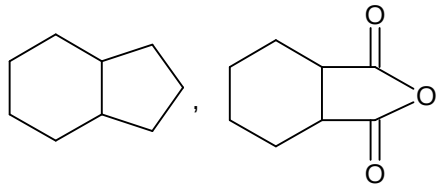
(D) 

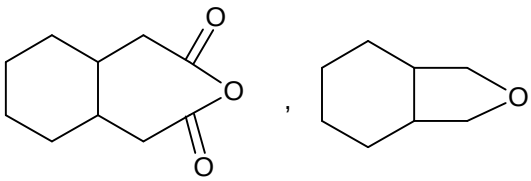
30.

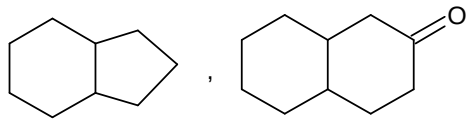


If P and Q are main products then P and Q are:

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Number of words can be formed using all the letters of the work 'HIPHIPHURRAY' in which all H's lie somewhere between R's is
   (A) (198)7! (B) (99)7!
   (C) (99)8! (D) (198)8!

2. Let a, b ($a \neq b$) are two non–zero complex numbers satisfying $|a^2 - b^2| = |\vec{a}^2 + \vec{b}^2 - 2\vec{a}\vec{b}|$ then
   (A) $\frac{a}{b}$ is purely real (B) $\frac{a}{b}$ is purely imaginary
   (C) $|\arg(a) - \arg(b)| = \pi$ (D) $|\arg(a) - \arg(b)| = \frac{\pi}{4}$

3. Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -K(T-t)$, where K > 0 is a constant and T is the total life in years of the equipment. Then the scrap value V(T) of the equipment is
   (A) $I - \frac{KT^2}{2}$ (B) $I - \frac{K(T-t)^2}{2}$
   (C) e^{-KT} (D) $T^2 - \frac{1}{K}$

Space for Rough work

4. If $\left(\tan^{-1}x\right)^2+\left(\cot^{-1}x\right)^2=\frac{5\pi^2}{8}$, then x is equal to

(A) –1 (B) 0
(C) 1 (D) none of these

5. If the equation $\left|\,|x-1|-6\lim\limits_{t\to\infty}\left(\frac{\sqrt{2t^2-t-1}-\sqrt{t^2-t+1}}{t\left(\tan\frac{\pi}{8}\right)}\right)\right|=k$ has four distinct solutions then the number of integral values of k is

(A) 4 (B) 5
(C) 6 (D) none of these

6. If $f(x+y)=f(x)+f(y)+|x|y+xy^2$, $\forall$ x, y $\in$ R and f′(0) = 0 then

(A) f(x) is twice differentiable at x = 0 (B) area bounded by f(x), x = 1, y = 0 is $\frac{1}{3}$
(C) f′(x) is continuous $\forall$ x $\in$ R (D) f′(x) is differentiable $\forall$ x $\in$ R

7. If f : [2, ∞[$\to$ [8, ∞[is a surjective function defined by $f(x)=x^2-(p-2)x+3p-2$, p $\in$ R then sum of values of 'p' is $m+\sqrt{n}$, m, n $\in$ N then m + n is equal to

(A) 20 (B) 10
(C) 30 (D) 40

8. For x, y $\in$ R, if $x^2-2xy+2y^2-6y+9=0$ then the value of (5x – 4y) is equal to

(A) 2 (B) 3
(C) 6 (D) 15

9. Let $f(x)=\begin{cases}2-x+a^2-9a-9, & x<2\\ 2x-3, & x\geq 2\end{cases}$ where a is a positive constant. It f(x) has local minimum at x = 2, then a lies in the interval

(A)]0, 1] (B) [10, ∞[
(C) [1, 10] (D) [2, 10[

Space for Rough work

10. If a, b $\in$ R and satisfy $a = b + \cfrac{1}{a + \cfrac{1}{b + \cfrac{1}{a + ...\infty}}}$, $b = a - \cfrac{1}{b + \cfrac{1}{a - \cfrac{1}{b + ...\infty}}}$ then $a^2 - b^2$ is equal to

(A) –1 (B) 2
(C) 1 (D) 0

11. The probability that a randomly selected calculator from a store is of brand 'r' is proportional to r(r = 1, 2, ….., 6). Further, the probability of a calculator of brand 'r' being defective is $\frac{7-r}{21}$, r = 1, 2, 3, 4, 5, 6. If probability that a calculator randomly selected from the store being defective is $\frac{p}{q}$, where p and q are co–prime then the value of (p + q) is

(A) 63 (B) 60
(C) 71 (D) 21

12. If P(r) be number of points (x, y), x, y $\in$ I which lie inside or on the curve $|xy| - r(|x| + |y|) + r^2 = 0$, r $\in$ N and if $\lim\limits_{n\to\infty}\left(\frac{\sum\limits_{r=1}^{n-1}\sqrt{P(r)}}{an^3 + bn^2 + c}\right) = \frac{1}{2}$ then (a + b), is

(A) 1 (B) 2
(C) 3 (D) 4

13. It $ax + b\sec(\tan^{-1} x) = c$ and $ay + b\sec(\tan^{-1} y) = c$, then $\frac{x+y}{1-xy}$ is equal to

(A) $\frac{2ac}{a-c}$ (B) $\frac{2ac}{a^2-c^2}$
(C) $\frac{a+c}{1-ac}$ (D) $\frac{ac}{a^2+c^2}$

Space for Rough work

14. Consider, $f(x)=\begin{cases}\dfrac{e^{(x-2)\ln\sqrt{5^x}+\frac{1}{3x}\log_2\left((4^x)^3\right)}-e^2}{(x-2)\tan x}, & x>0\\ \lambda\dfrac{\left((\sqrt{5})^x-1\right)\left(e^{x^2+2}-e^2\right)}{(2\sin x-\sin 2x)}, & x<0\end{cases}$. If $\lim\limits_{x\to 0} f(x)$ exists then λ is equal to

(A) 1 (B) –1
(C) 2 (D) ln 5

15. The absolute value of the expression $\left(\dfrac{1+\sqrt{3}\tan 1^o}{\sqrt{3}+\tan 1^o}\right)\left(\dfrac{1-\sqrt{3}\tan 1^o}{\sqrt{3}-\tan 1^o}\right)\dfrac{\tan 89^o}{\cot 3^o}$ is equal to

(A) 0 (B) 1
(C) 2 (D) 3

16. The value of $\lim\limits_{x\to 0}\dfrac{1}{x}\int\limits_0^x(1+\sin 2t)^{1/t}\,dt$ is equal to

(A) 2 (B) 1
(C) e (D) e^2

17. Let f be a polynomial function such that for all real x, $f(x^2+1)=x^4+5x^2+2$, then $\int f(x)dx$ is

(A) $\dfrac{x^3}{3}+\dfrac{3x^2}{2}-2x+c$ (B) $\dfrac{x^3}{3}+\dfrac{3x^2}{2}+2x+c$

(C) $\dfrac{x^3}{3}-\dfrac{3x^2}{2}-2x+c$ (D) $\dfrac{x^3}{3}-\dfrac{3x^2}{2}+2x+c$

18. If $f(x)=(x-1)^4(x-2)^3(x-3)^2(x-4)$, then the value of $(f'''(1)+f''(2)+f'(3)+f'(4))$ is equal to

(A) 638 (B) $3(6^3)$
(C) 50 (D) 0

Space for Rough work

19. The first two terms of a geometric progression add up to 12. The sum of third and fourth terms is 48. If the terms are alternately positive and negative, then
(A) common ratio of G.P. is –2 (B) common ratio of G.P. is –3
(C) sixth term of G.P. is 128 (D) none of these

20. If P(x, y, z) is a point on the line segment joining Q(2, 3, 4) and R(3, 5, 6) such that the projection of the vector $\overrightarrow{OP}$ on the axes are $\frac{13}{5}, \frac{21}{5}, \frac{26}{5}$ respectively. The P divides QR in the ratio
(A) 2 : 3 (B) 3 : 1
(C) 1 : 3 (D) 3 : 2

21. Area of triangle whose vertices are (a, a^2), (b, b^2), (c, c^2) is $\frac{1}{2}$, and area of another triangle whose vertices are (p, p^2), (q, q^2) and (r, r^2) is 4, then the value of $\begin{vmatrix} (1+ap)^2 & (1+bp)^2 & (1+cp)^2 \\ (1+aq)^2 & (1+bq)^2 & (1+cq)^2 \\ (1+ar)^2 & (1+br)^2 & (1+cr)^2 \end{vmatrix}$, is
(A) 2 (B) 4
(C) 8 (D) 16

22. The line parallel to the x–axis and passing through the intersection of the lines ax + 2by + 3b = 0 and bx – 2ay – 3a = 0, where $(a, b) \neq (0, 0)$ is
(A) below the x–axis at a distance of $\frac{3}{2}$ from it (B) below the x–axis at a distance of $\frac{2}{3}$ from it
(C) above the x–axis at a distance of $\frac{3}{2}$ from it (D) above the x–axis of distance of $\frac{2}{3}$ from it

23. The locus of feet of perpendicular from either foci of the ellipse $(x – y + 1)^2 + (2x + 2y – 6)^2 = 20$ on any tangent will be
(A) $x^2 + y^2 + 2x + 4y + 5 = 0$ (B) $x^2 + y^2 + 2x + 4y – 5 = 0$
(C) $x^2 + y^2 – 2x – 4y – 5 = 0$ (D) $x^2 + y^2 – 2x – 4y + 5 = 0$

24. For $x \neq \frac{n\pi}{2}$, where $n \in I$, the range of function
$f(x) = \text{sgn}(\sin x) + \text{sgn}(\cos x) + \text{sgn}(\tan x) + \text{sgn}(\cot x)$ is
(A) {–2, 4} (B) {–2, 0, 4}
(C) {–4, 0, –2, 4} (D) {0, 2, 4}

Space for Rough work

25. If $\cos\theta = \frac{a}{b+c}$, $\cos\phi = \frac{b}{a+c}$, $\cos\psi = \frac{c}{a+c}$, where $\theta, \phi, \psi \in (0, \pi)$ and a, b, c are sides of triangle ABC then $\tan^2\frac{\theta}{2} + \tan^2\frac{\phi}{2} + \tan^2\frac{\psi}{2}$ is equal to

(A) 1 (B) 2
(C) 3 (D) 4

26. Let A = {1, 2, 3, 4}. The number of different ordered pairs (B, C) that can be formed such that $B \subseteq A$, $C \subseteq A$ and $B \cap C = \phi$

(A) 3 (B) 9
(C) 27 (D) 81

27. If $\cos^2 x - (c - 1)\cos x + 2c \geq 6$ for every $x \in R$, then the true set of values of c is

(A) $[4, \infty[$ (B) $[2, \infty[$
(C) $]-\infty, -2]$ (D) $]-\infty, -4]$

28. If the equations $px^2 + qx + r = 0$ and $rx^2 + qx + p = 0$ $(p \neq r \neq 0)$ have a negative common root, then the value of $(p - q + r)$ is equal to

(A) –1 (B) 1
(C) 2 (D) 0

29. If $f(x) = \left([\{x\}]\tan^{-1}\left(\frac{x^2 - 3x - 1}{x^2 - 3x + 5}\right) + 3 - x^7\right)^{\frac{1}{7}}$. Where [k] and {k} denotes greatest integer and fractional part functions of k respectively, then the value of $f^{-1}(50) - f(50) + f(f(100))$, is

(A) 0 (B) 25
(C) 50 (D) 100

30. In triangle ABC, if $a \sin A \sin B + b \cos^2 A = \sqrt{2}a$, then the value of $\left(\frac{b}{a}\right)$ is

(A) $\frac{1}{2}$ (B) 2
(C) $\sqrt{2}$ (D) 1

Space for rough work